The Political Economy of Resilience: Climate Information and Bureaucratic Effectiveness

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Abstract

This project will explore the political economy of government effectiveness and climate-resilient policymaking. We use a randomized field experiment in Turkana County, Kenya to test whether provision of climate information to government officials leads to better-informed officials and better policymaking in the face of climate shocks. We also explore more generally how information flows within bureaucratic hierarchies, and how bureaucratic incentives shape decisionmaking and communication.

Keywords

Climate adaptation, climate information, pastoralism, climate shocks, Kenya

JEL Codes

Study Pre-Registration

Study Pre-Registered on AEA RCT Registry as AEARCTR-0018628. https://www.socialscienceregistry.org/trials/18628

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1 Introduction

Governments and bureaucracies in arid and semi-arid regions face a wave of coping and adaptation challenges as climate change leads to more extreme and frequent shocks. These challenges are especially severe as future shocks may be dissimilar from past incidents they have faces, and policymaker and policy implementers will need to respond to new issues. This makes a well-informed bureaucracy crucial for effective adaptation in order to both identify the correct response to any given situation, as well as evaluating the relative severity of different to allocate limited coping resources across different threats. The urgency of creating evidence to guide decisionmaking is widely acknowledged, and a significant research resources have been directed at creating the knowledge that government officials need. Despite the *existence* of this knowledge, its *effect* is limited by access, and the costs for any one official to seek out and acquire evidence-based policy guidance. In this project we evaluate a novel unified Climate Information System (CIS) in Turkana County that provides a single stremlined point of access where research projects relevant to Turkana can easily share problem-oriented knowledge outputs tailored to the needs of government officials.

As the growing literature on states and bureaucracies has made clear, better information may not necessarily lead to better policy in the presence of a range of pernicious principal-agent problems. (e.g., Besley et al. (2022), Bobonis et al. (2016), Besley (2007)) In our evaluation of this Climate Information System, we will aim to uncover the political economy constraints on climate-informed policymaking. In particular, we will explore the dynamics of information between frontline policy implementers and their superiors in the bureaucratic hierarchy. In addition to being important for this particular context, these dynamics are a general concern for the efficient production of public goods.

2 Literature Review

This paper contributes to several interconnected literatures in economics and political science on bureaucratic effectiveness, the role of information in government, and the political economy of hierarchical organizations. This is a fast growing area of research of great importance to effective policymaking for development, however, the way information is used in decisionmaking and increasing effectiveness has not been explored.

A growing body of research has established that the quality of bureaucracies is a fundamental determinant of state capacity and economic development. Besley and Persson (2013) the authors provide empirical evidence of this relationship, empirically link state capacity, including the ability to collect information and implement policy, to long-run economic performance. More recently, Besley et al. (2022) survey the economics of bureaucracy, emphasizing that bureaucratic organizations face distinctive challenges of selection, incentives, and information aggregation that shape policy outcomes. A key finding from literature is that well-intentioned governments may still fail to deliver effective policy if their bureaucracies lack the information or capacity to do so.

Empirical work has increasingly focused on identifying the specific mechanisms through which bureaucratic quality affects service delivery. Rasul and Rogger (2018) use data from the Nigerian civil service to show that management practices within bureaucracies,

including information management and monitoring, are strongly associated with public service delivery outcomes. Similarly, Best et al. (2023) demonstrate that individual bureaucrats matter for policy implementation in Brazil, with significant variation in effectiveness across officials even within the same organization. These findings underscore that bureaucratic effectiveness is not merely a function of institutional design (i.e. checks and balances or monitoring) but also of the information environment in which officials operate.

A key insight from the literature on organizations is that information is central to bureaucratic performance, but its acquisition and use are shaped by incentive structures and cognitive constraints.

The challenge of translating information into action within bureaucracies has been formalized theoretically. Aghion and Tirole (1997) model the allocation of authority within organizations, showing that the value of delegating decisions to subordinates depends critically on the quality of information at different levels of the hierarchy. When frontline officials have better local information but superiors have better strategic information, the optimal organizational structure depends on how easily information can be communicated and verified. Dessein (2002) extends this analysis, demonstrating that strategic communication between hierarchical levels is often distorted, and that more informative communication may counterintuitively reduce a superior's willingness to delegate.

The political economy of superior-subordinate relationships within bureaucracies is shaped by principal-agent problems that can undermine the effective use of information. Besley (2007) develop a model of political agency in which the quality of governance depends on the selection of politicians and bureaucrats, and the information available to citizens to hold them accountable. Research in this field generally focuses on information in the sense of monitoring and audits to punish or prevent improper behaviour. Bobonis et al. (2016) provide experimental evidence from Puerto Rico showing that the threat of audits improved municipal fiscal management, but that these effects depended on the political incentives of incumbents. This shows that information, specifically public information about the activities of politicians, interacts with political economy constraints in shaping bureaucratic behavior. Within bureaucratic hierarchies specifically, several studies have documented how information asymmetries between superiors and subordinates distort policy implementation. Khan et al. (2019) study tax inspectors in Pakistan and find that providing performance information to both inspectors and their supervisors significantly increased tax collection, but only when the information was made available to supervisors as well, highlighting the importance of vertical information flows within hierarchies. Ashraf et al. (2020) provide evidence from Zambia's health sector that the introduction of monitoring and information systems can have unintended consequences when they interact with existing hierarchical dynamics. Specifically, they find that community health workers who were subjected to greater monitoring by superiors were less intrinsically motivated, suggesting that the effect of information provision depends on how it alters the relational dynamics between levels of the hierarchy. Relatedly, Muralidharan et al. (2016) show in the context of India's workfare program that biometric smartcards (which improved the flow of information about payments) reduced leakage and improved outcomes for beneficiaries, in part by reducing the discretion available to intermediaries in the bureaucratic chain.

The role of information in enabling climate adaptation by governments remains comparatively understudied, though several recent contributions are relevant. Dell et al. (2014) review the broader literature on the economic impacts of climate, noting that government effectiveness in responding to climate shocks varies enormously and is a key determinant of welfare outcomes.

The provision of climate information to households and communities has been studied more extensively. **?** show that Indian farmers who received and acted on monsoon forecasts adjusted their crop choices and experienced significant income gains. **?** demonstrates that improved weather forecasts in the United States led to meaningful reductions in mortality from extreme temperatures, with effects concentrated among populations with better access to information. However, these studies focus primarily on end-users rather than on the government officials responsible for translating climate information into policy. The bureaucratic intermediation of climate information (how officials receive, interpret, and act upon forecasts within hierarchical organizations) remains a significant gap in the literature.

For this project we will rely on the subjective evaluation of proposed responses to evaluate the quality of decisionmaking by government officials. To do this, we will follow best practice in the literature on subjective evaluation of bureaucrats. This includes work such as In Rasul and Rogger (2018) the authors code administrative project reports to evaluate the effect of management practices on the successful completion of projects. In another paper, Limodio (2021) uses internal project performance ratings of the World Bank to study the allocation of World Bank staff Similarly, Bertrand et al. (2020) who use a survey to collect subjective assessments of senior Indian bureaucrats.

3 Context

This project is being implemented in Turkana County, Northern Kenya. Turkana is a hot and arid setting, with a temperature range between 21 °C and 40 °C and a mean temperature of 30 °C. Turkana County has suffered from severe climate shocks in recent years, including droughts, flash floods, extreme heat, among other hazards. The responses to these hazards are organized at several layers of government, with local government involved both in implementing policy and communicating risks and opinions on optimal responses up the hierarchy to leadership of Turkana County Government and the various relevant ministries. These local officers are often poorly informed about potential risks and modelling of different hazards, despite the fact that there is a wealth of climate information produced, both by government meteorological agencies as well as by research projects active in the region and data collected by other local stakeholders such as NGOs.

4 Research Design

In this project we will implement a randomized experiment conducted across the staff of the Turkana County Government (TCG). This full roster of employees includes approximately 5,000 officials at the local level across ministires. Our main sample will be the local Ward Climate Change Committees, who represent the frontline layer of TCG

officials, who make decisions themselves and provide information and recommendations that feeds into other decisions. While these WCCCs are the main focus, as they are the main local unit of government emergency response, we will also include other key members of the TCG administration to understand whether other layers of government would also be able to benefit from improved information, while also understanding where the best information resides within the hierarchy.

In partnership with the office of the Turkana County Secretary and Governor of Turkana, we will send out a questionnaire asking government officials to propose a response to a set of hypothetical scenarios. They are allowed to reference whatever sources they have at their disposal to guide their answer, and we will - to the best of our ability - ensure that this is not overtly related to the CIS intervention. We then test whether a) CIS-sensitized people actually mention/use the information, b) whether they choose better policy, and c) whether they answer factual questions about current conditions and forecasts more accurately. We code up and test for mention of climate info (are they more likely to use info) and then "score" the response to test whether they picked the right approach (did CIS make policy better).

Our research project is therefore designed to answer two broad research questions. The first is the major focus of this Registered Report, and the second is a more exploratory endeavour. The research questions are outlined in more detail below where we outline the hypotheses and data used to test them.

- **RQ1**: Does onboarding to an online integrated Climate Information Platform (CIS) meaningfully improve how well-informed government officials are about climate conditions and risks?

- **RQ2**: Does this improved information lead to better choices in response to climate hazards?

4.1 Experimental Sample

The eligible population for this study will be the employees of the Turkana County Government, with a focus on the Ward Climate Change Committees (WCCCs) as the main unit of analysis. The WCCCs are responsible for local-level climate change adaptation and disaster response planning, making them a critical target for information interventions. We will also include other key administrative staff across various ministries to understand information within the bureaucracy. While WCCCs are the main focus, decisionmaking about disaster response is distributed throughout the TCG hierarchy and other government actors also play an important role in climate resilience so understanding their information and decisionmaking is also crucial.

4.2 Descriptive Statistics and Balance

Baseline data collection has not yet begun, so we cannot yet present descriptive statistics. We will conduct balance tests using regressions of the form

$$\text{AnyTreatment}_{iv} = \beta_0 + \mathbf{\Gamma}_{iv} + \varepsilon$$

and then testing for the joint significance of the vector of baseline variables $\mathbf{\Gamma}_{iv}$. We will test this using linear regression models, logistic models, with/without strata fixed effects, and using randomization inference tests as well as frequentist for both the bundled comparison of all climate information groups against the control and for the two treatment arms against each other.

4.3 Hypotheses

4.3.1 Theory of Change

The theory of change, and assumptions, underlying our policy innovation follows.

1. The first assumption is that TCG officials lack simple access to high quality climate forecasts and other climate-relevant information.

2. First, this intervention is easy to access and understand for frontline government officers, and therefore they will actually access it and absorb the information provided by the CIS.

3. Second, TCG officials will choose to use this information when they make decisions about how to respond to climate hazards.

4. Third, finally, this additional information will lead to government officers choosing better responses to climate risks their communities face.

Main Hypotheses

1. Hypothesis 1: Levels of information among untreated TCG officials is low. This is not so much a hypothesis to test, demonstrate with descriptive stats.

2. Hypothesis 2: Government officials with access to the CIS will actually choose to access the CIS and will be better able to answer questions about current conditions more accurately and their expectations about seasonal conditions will be closer to the forecasts provied on the CIS.

3. Hypothesis 3: When government officials respond to a climate shock, they decide to access the CIS system to check the evidence / information as part of their planning.

4. Hypothesis 4: When government officials respond to a climate shock, they choose better responses to the shock.

5. Hypothesis 5: Hierarchical dynamics mean that government officials will change their responses when they know that their response will be shared with a superior, and this change will be in the direction of less costly responses, more positive spillovers to other communities, and more alignment with stated government policy.

4.4 Methodological Framework

The overall methodology here is a clustered RCT, with treatment assigned at the WCCC level and at the office level for other TCG actors. Randomization will be stratified at the level of sub-counties to ensure balance across climate conditions.

4.5 Intervention

The intervention in this project will be access and onboarding to an integrated Climate Information System (CIS) platform.[1] This platform will provide a wide range of climate-relevant information including, but not limited to, current condition of vegetation, livestock health, borehold functionality and recent incidents of migration and conflict. This will also include a range of forecasts of seasonal conditions including Kenya Meteorological Services Authority (KMSA, formerly KMD) rainfall, temperature, and flood risk forecasts as well as forecasts of healthcare demand surges (e.g. a spike in malnutrition cases).

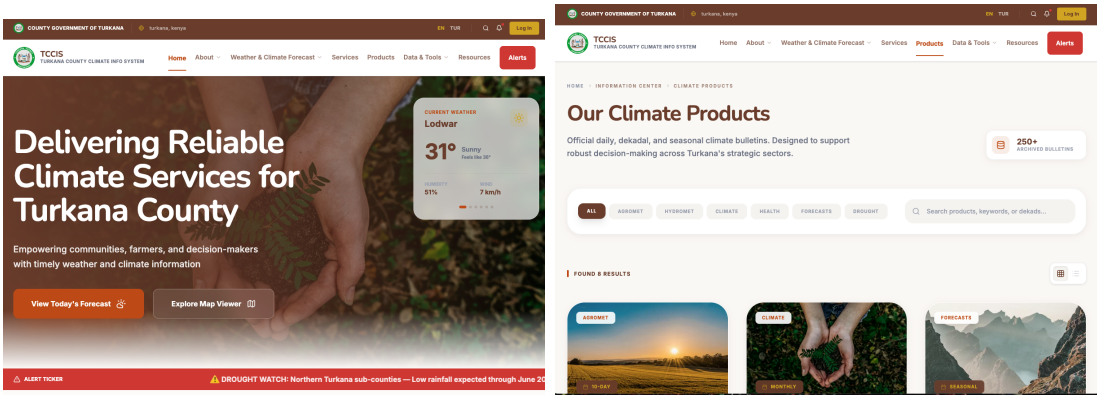


Figure 1: Screenshots of Current Prototype CIS

This treatment will be delivered to the government officials in the treatment group by enabling their online access to the platform, and an in-person workshop explaining how to use it and the information available to them through the platform.

4.6 Timeline

The overall timeline for this experiment is as follows, and is also presented in Figure **??**. The timing is mostly decided by the March-April-May rainfall season, and when the KMD produces their climate information.

- June 2026: Prepare first viable working prototype of CIS, and pilot the data collection instrument for measuring readiness and the panel scoring methodology. Collect baseline data if time allows.
- July 2026: Begin rolling out CIS to the treatment group

[1] Click here for the current prototype system.

- August 2026: Data Collection and scoring by panel.
- September 2026: Data analysis, write-up of results.

5 Data

The main outcome data in this project is collected using online surveys implemented using the Qualtrics survey platform.

We will collect two main types of data. The first is a free-text response to a hypothetical climate hazard storyline that could potentially impact their constituency / community. We will ask them to explain how they would respond to this shock, step-by-step. Our intention is for this to be as open-ended and unstructured as possible, so as not to prime respondents, but after piloting we may add follow-up or sub-questions that focus the data collected. We will repeat this exercise for multiple hypothetical scenarios, and we will randomize the order of these scenarios to avoid ordering effects.

After the collection of this free-text response question, we will ask questions about the current climate conditions in their region, and their perceptions over what will happen this season.

We will also tell participants that we would like to share one of their randomly-selected responses to be shared (anonymously) with another officer further up the hierarchy. We will offer them the chance to revise their responses to see if hierarchy effects and relationships with superiors lead to self-censorship or systematically different responses.

5.1 Data collection and processing

Refer to appendix 7 for the exact questionnaire used to construct these outcomes.

1. Descriptive Analysis of CIS Use
   (a) Average number of logins and time spent on platform
   (b) Type of information accessed (e.g. forecasts, current conditions, etc.)
2. Use of CIS information in decisionmaking
   (a) Mentions of CIS information in free-text responses to hypothetical scenarios
3. Accuracy of beliefs
   (a) Accuracy of beliefs about current conditions (e.g. vegetation, livestock health, etc.)
   (b) Accuracy of beliefs about seasonal forecasts (e.g. rainfall, temperature, etc.)
4. Quality of responses
   (a) Scoring of free-text responses to hypothetical scenarios by an expert panel, with scores based on the quality of the response and whether it is consistent with the information provided on the CIS.

(b) Breakdwon of scoring across criteria such as: equity considerations, cost-effectiveness, etc.

5. Change in responses when shared with superior
   (a) Estimated cost of responses
   (b) Spillovers to other regions / communities nearby
   (c) Alignment with stated government policy

5.2 Data Collection Timeline

We will first collect pilot data followed by baseline data in June 2026. We will then roll out the intervention in July 2026, and collect endline data in August 2026. We will then analyze the data and write up results in September 2026.

6 Analysis

6.1 Statistical methods

This project will be analyzed using standard empirical methods for RCTs, with regression analysis carried out in Stata. For all regressions we will report standard significance tests as well as randomization inference p-values.

6.2 Empirical Specification / Statistical model

The empirical model for most of the main outcomes (manipulation check, expectation deviation from forecasts, and confidence in forecasts, livelihood outcomes) is as follows:

$$Y_{i,o,e} = \beta_0 + \beta_1 \cdot \text{CIS}_{i,o} + Y_{i,o,b} + \varepsilon_{i,o} \tag{1}$$

where $\text{CIS}_{i,o}$ is an indicator variable for the official i in office o having been randomly assigned to receive access to the Climate Information System, and $Y_{i,o}$ is the baseline value of the endline outcome. Errors $\varepsilon_{i,o}$ are clustered at the office level, the unit of assignment.

6.3 Statistical Power

We will analyze statistical power once baseline/pilot data has been collected, and we are able to approximate the variance of the outcomes as well as intracluster correlation.

6.4 Multiple outcome and multiple hypothesis testing

We will minimize multiple hypothesis testing by creating a single weighted index for each of the 6 main outcomes in Section 5.1, where for intended and actual behaviours, the signs of actions will be flipped so that the index overall can be interpreted as a higher number being more good-state actions and fewer bad-state actions. For these six outcomes we will compute standard errors by using stepdwon FWER and FDR p-values.

6.5 Heterogeneous Effects

For heterogeneous effects, we will use the data driven Conditional Average Treatment Effect (CATE) methodology as in Athey and Imbens (2016), Athey and Wager (2018), Athey and Wager (2019).

7 Interpreting Results

If the coefficient for $\text{CIS}_{i,o}$ in equation 1 is positive and significant for outcome 2, we interpret this as evidence that onboarding to the CIS led to actual use of climate information in the day-to-day activities of government officers, as part of their self-reported response to threats.

If the coefficient for $\text{CIS}_{i,o}$ in equation 1 is positive and significant for outcome 3, we interpret this as evidence that access to the Climate Information System had a meaningful effect on the knowledge of government officials about the conditions in their area as well as better awareness of forecasts made by various meteorological and scientific teams.

For outcome 4, if the coefficient for $\text{CIS}_{i,o}$ is positive and significant, we interpret this as evidence that the CIS led to better decisionmaking by government officials, which indicates greater capacity to adapt to climate shocks. While this may not translate into better actual outcomes, it is an important first step in the causal chain to understand if poor decisionmaking is a barrier to effective policy implementation that can be easily improved.

Finally, for outcome 5, we interpret the treatment effect as the effect of bureaucratic hierarchies on decisionmaking. We expect that hierarchies will lead to more low-cost responses, proposals that may have more consideratio nfor welfare of other nearby communities, and are closer to stated government policy. We would interpret this result as evidence that, while frontline officers may be best informed, superiors in hierarchies are perceived to have incentives to consider optimal decisionmaking at a larger big-picture scale.

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Questionnaire - TCG Officials

Information and Consent

Participant Information Sheet

Researcher names: Julian Dyer, Dennis Ochieng, Ellen Dyer, Peter Opeyemi James, and Dalmas Omia

Title: Turkana County Climate Readiness Drill

You are being invited to take part in this exercise, the Turkana County Climate Readiness Drill, is to understand how government officials in Turkana respond to climate shocks. It is important you understand why this research is being done and what your participation will involve.

Purpose of the research:
The purpose of this phase of the research project is to learn more about how government officials respond to climate shocks in Turkana County. We want to understand how you would go about responding to various challenges your community might face.

Who is doing this research?
The lead researcher of this study is Dr. Julian Dyer, a Lecturer in the Department of Economics at the University of Exeter. The other researchers are Peter Opeyemi James from the University of Exeter, Dr. Ellen Dyer the University of Oxford, and Dr. Dalmas Omia and Dennis Ochieng from the University of Nairobi.

Why have I been approached?
You have been approached because you are a member of the Turkana County Government's administrative system.

What would taking part involve?
If you choose to participate, you will fill out a short survey asking about your recommended approach to responding to climate shocks in Turkana, as well as asking about your perception of current and likely future conditions. This data will be collected on the Qualtrics survey platform and you can view their Data Protection and Privacy policies here: https://www.qualtrics.com/support/survey-platform/getting-started/data-protection-privacy/

What are the possible benefits of taking part?
The benefit of taking part is making an important contribution to informing how the Turkana County Government responds to climate shocks. This may help improve resilience of Turkana with benefits for the community.

What are the possible disadvantages and risks of taking part?
Risks of participation in this study are minimal. There will be no physical discomfort and minimal mental discomfort.

Voluntary participation and your right to withdraw:
Your participation in this survey is entirely voluntary. You are under no obligation to participate and you can choose to stop at any time. If you do not want to carry on you can choose to end the survey at any point and if you stop before completing the survey your information will not be saved.

If you choose not to participate once the interview has been completed, please contact any member of the research team and your information will be removed from the data and from any future publications. Once data has been shared (without identifying information) as part of the publication process, we will not be able to delete your data from that dataset. Once data has been anonymised and personally identifiable information deleted (which will happen within three years of the project ending) then it will not be possible to identify your responses to delete them.

How will my information be kept confidential?
Your personally identifying information will not be shared, so your information will remain confidential. We will be storing all data collected as part of this project on a secure site accessible only to team members. Any personally identifiable data will be destroyed within 3 years of the completion of the project. We will keep your identifiable contact information for up to three years of project completion in order to contact you for a potential follow-up.

What will happen to the results of this study?
The results of the interviews will be shared with the research team and the researchers at the University of Exeter, the University of Nairobi, and the University of Oxford. This data (without your personally identifiable information) will be analyzed

for an academic publication. Results of this study (without any information that could identify you individually) will be shared as part of this academic publication and policy briefings. The results will be published and presented in academic papers and seminars as well as with policymakers in Turkana. If you want to follow up with this research, please visit the lead researcher's (Julian Dyer's) website or feel free to contact the researcher directly.

Who is organising and funding this study?
The sponsor for this phase of the study is the IDRC / FCDO Research For Impact initiative.

Who has reviewed this study?
This project has been reviewed by the Business School Research Ethics Committee at the University of Exeter (Application Number 13228393).

Contact for any questions or requests regarding your participation in this research:
In the event of queries or requests you may contact me using the following contact information.
Please email *j.dyer3@exeter.ac.uk*

To contact the Research Ethics Committee please email Co-Directors Dr. Steven Boyne (S.Boyne@exeter.ac.uk) and Dr. Helena Fornwagner (H.Fornwagner@exeter.ac.uk)

You can also contact the University Research Ethics and Governance Team if you wish to make a complaint or comment please email cgr-reg@exeter.ac.uk,

You can also contact members of the research team:
Julian Dyer (j.dyer3@exeter.ac.uk)
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Dennis Ochieng (denniso@uonbi.ac.ke)
Peter Opeyemi James (pj324@exeter.ac.uk)
Dalmas Omia (ochiengomia@uonbi.ac.ke)

Thank you for your interest in this project.

Block 1 - Recommended Policies

CONSENT

1. I confirm that I have read the information provided above.

2. I confirm that I am 18 years old or older.

3. I understand that my participation is voluntary and that I am free to withdraw at any time.

4. I understand that relevant sections of the data collected during the study may be looked at by members of the research team and individuals from the University of Exeter where it is relevant to my taking part in this research. My responses will also be anonymously reviewed by a panel of experts. I give permission for these individuals to have access to my records.

5. I understand that my contact details will be kept for up to three years after the project is completed in case the research team want to contact me for a follow-up survey.

6. I understand that taking part involves survey data to be collected and used for the purposes of anonymized inclusion in academic publications, project publication/website, policy briefings, and further grant applications.

7. I agree to take part in the above project.

○ I confirm that I have read the consent statements and agree to participate in this study.
○ I do not consent to agree to participate in this study.

If you had a budget of 500,000 KES to spend on preparation for climate hazards in your ward/ community this season, what would you prioritize? Why?

This season, an NGO approaches you and says they have either a feeding program or a livestock vaccination project but can only implement one in your area. Which would you prefer? Why?

- ○ Feeding program
- ○ Livestock vaccination
- ○ Not sure

Please explain your answer to the previous question and explain exactly what you considered while making your decision.

If you had to recommend one policy to NGO stakeholders that they should implement this season, what would it be?

We will share the aggregated (non-identifiable) results with NGO stakeholders, so please consider this carefully as your insights may help influence policy decisions.

- ○ Water delivery
- ○ Borehole
- ○ Food aid
- ○ Agroforestry program
- ○ Small cash handouts to households
- ○ Road repairs / market development
- ○ Quick low-capital business development (e.g. motorbikes on loan/installment)
- ○ High-capital long-term business development (e.g. agricultural value addition chain)

We will now ask you a number of questions related to climate shocks that might occur in your area of Turkana and how the government should respond to them. We will give you 3 scenarios that might occur, and we ask that you explain the policy response that would be best in each situation. Please be as detailed as possible and tell us about each stage of your response, from who you would consult, your fact-finding process and the factors that go into each decision you would make as part of this proposed response. Your answers will be anonymously reviewed by a panel of experts and will help inform Turkana's preparations for climate hazards, so please give these responses careful consideration.

Even if you do not make these decisions yourself, respond AS IF you were making the decision

or if you were providing your recommendation to the person that does make the decision.

Scenario 1:

"It is late March and the long rains have not yet begun. People in your community are concerned that the season may fail and some have begun talking of long migrations into areas near the Pokot border and possibly into South Sudan."

How do you think the county government should respond to this scenario? What would be the proposed action you suggest to the government? Please be as detailed as possible, explain all the different resources you would draw on, and explain all the crucial decisions you make to arrive at your proposed action the government should take to respond to this situation.

Scenario 2:

"At a community baraza, a number of people mentioned that the river appears to be running unusually high for this time of year. They think there may be flooding this season and have brought their concerns to you."

How do you think the county government should respond to this scenario? What would be the proposed action you suggest to the government? Please be as detailed as possible, explain all the different resources you would draw on, and explain all the crucial decisions you make to arrive at your proposed action the government should take to respond to this situation.

Scenario 3:

"It is early in the hot season and temperatures have been unusually high for several consecutive days. Health workers report that elderly residents, young children, and livestock herders are falling ill, and some community members say the heat is worse than they can remember. People are asking what the government will do to help them cope."

How do you think the county government should respond to this scenario? What would be the proposed action you suggest to the government? Please be as detailed as possible, explain all the different

resources you would draw on, and explain all the crucial decisions you make to arrive at your proposed action the government should take to respond to this situation.

If you had to recommend one policy to decisionmakers in Turkana County Government that should be implemented this season, what would it be?

We will share the aggregated (non-identifiable) results with Turkana County Government stakeholders, so please consider this carefully as your insights may help influence policy decisions.

- ○ Water delivery
- ○ Borehole
- ○ Food aid
- ○ Agroforestry program
- ○ Small cash handouts to households
- ○ Road repairs / market development
- ○ Quick low-capital business development (e.g. motorbikes on loan/installment)
- ○ High-capital long-term business development (e.g. agricultural value addition chain)

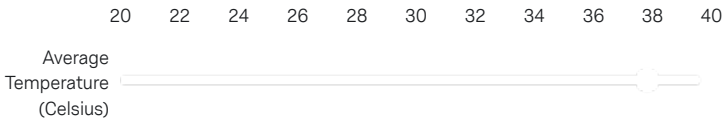
Block 2 - Current Conditions

We would now like to ask what you think the current conditions are in your ward.

1. What is the current vegetation condition?

- ○ Poor
- ○ Normal
- ○ Good

What has the average temperature been over the last month?



How many days has it rained over the last month? (Write number in box below).

Block 3 - Expectations for next month

We would now like to ask what you think the conditions ***will be next month*** in your ward.

1. What do you think the vegetation condition will be one month from now?

- ○ Poor
- ○ Normal
- ○ Good

2. What do you expect the average temperature to be over the next month?

- ○ 25-28 Degrees Celsius
- ○ 29-30 Degrees Celsius
- ○ 30-32 Degrees Celsius
- ○ 32-34 Degrees Celsius

3. How many days of rain do you expect over the next month?

- ○ 0 to 4
- ○ 5 to 9
- ○ 10 to 14
- ○ 15 to 20
- ○ 21 to 25
- ○ 25+

4
How much do you expect it to rain over the next month?

- ○ Below Average
- ○ Average
- ○ Above Average

Feedback

Thank you for participating in this exercise. You have now finished the survey.

If you have any comments, questions, or feedback about the study or your participation, please share them in the box below and feel free to contact researchers by email:

Julian Dyer (j.dyer3@exeter.ac.uk)
Ellen Dyer (ellen.dyer@ouce.ox.ac.uk)
Dennis Ochieng (denniso@uonbi.ac.ke)
Peter Opeyemi James (pj324@exeter.ac.uk)
Dalmas Omia (ochiengomia@uonbi.ac.ke)

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Questionnaire - Expert Review

Information and Consent

Participant Information Sheet

Researcher names: Julian Dyer, Dennis Ochieng, Ellen Dyer, Peter Opeyemi James, and Dalmas Omia

Title: Turkana County Climate Readiness Drill

You are being invited to take part in this exercise, the Turkana County Climate Readiness Drill, is to understand how government officials in Turkana respond to climate shocks. It is important you understand why this research is being done and what your participation will involve.

Purpose of the research:
The purpose of this phase of the research project is to learn more about how government officials respond to climate shocks in Turkana County. We want to understand how you would go about responding to various challenges your community might face.

Who is doing this research?
The lead researcher of this study is Dr. Julian Dyer, a Lecturer in the Department of Economics at the University of Exeter. The other researchers are Peter Opeyemi James from the University of Exeter, Dr. Ellen Dyer the University of Oxford, and Dr. Dalmas Omia and Dennis Ochieng from the University of Nairobi.

Why have I been approached?
You have been approached because you are an expert in issues related to climate change, coping strategies, and adaptation measures in Turkana or Kenya more broadly.

What would taking part involve?
If you choose to participate, you will fill out a short survey where we ask you to review free-text responses given by members of Turkana County Government. These questions asked them about their recommended approach to responding to climate shocks in Turkana, and proposals for how to use adaptation funding. We will provide you with a rubric and examples to help your scoring. This data will be collected on the Qualtrics survey platform and you can view their Data Protection and Privacy policies here: https://www.qualtrics.com/support/survey-platform/getting-started/data-protection-privacy/

What are the possible benefits of taking part?
The benefit of taking part is making an important contribution to measuring the capacity of the Turkana County Government to respond well to climate shocks. This may help improve resilience of Turkana with benefits for the community. This is a novel approach that may be scaled up to other settings relevant to your own interests.

What are the possible disadvantages and risks of taking part?
Risks of participation in this study are minimal. There will be no physical discomfort and minimal mental discomfort.

Voluntary participation and your right to withdraw:
Your participation in this survey is entirely voluntary. You are under no obligation to participate and you can choose to stop at any time. If you do not want to carry on you can choose to end the survey at any point and if you stop before completing the survey your information will not be saved.

If you choose not to participate once the interview has been completed, please contact any member of the research team and your information will be removed from the data and from any future publications. Once data has been shared (without identifying information) as part of the publication process, we will not be able to delete your data from that dataset.

How will my information be kept confidential?
Your personally identifying information will not be shared, so your information will remain confidential. We will be storing all data collected as part of this project on a secure site accessible only to team members. Any personally identifiable data will be destroyed within 3 months of the data collection.

What will happen to the results of this study?
The results of the interviews will be shared with the research team and the researchers at the University of Exeter, the University of Nairobi, and the University of Oxford. This data (without your personally identifiable information) will be analyzed

for an academic publication. Results of this study (without any information that could identify you individually) will be shared as part of this academic publication. The results will be published and presented in academic papers and seminars. If you want to follow up with this research, please visit the lead researcher's (Julian Dyer's) website or feel free to contact the researcher directly.

Who is organising and funding this study?
The sponsor for this phase of the study is the IDRC / FCDO Research For Impact initiative.

Who has reviewed this study?
This project has been reviewed by the Business School Research Ethics Committee at the University of Exeter (Application Number 13228393).

Contact for any questions or requests regarding your participation in this research:
In the event of queries or requests you may contact me using the following contact information.
Please email *j.dyer3@exeter.ac.uk*

To contact the Research Ethics Committee please email Co-Directors Dr. Steven Boyne (S.Boyne@exeter.ac.uk) and Dr. Helena Fornwagner (H.Fornwagner@exeter.ac.uk)

You can also contact the University Research Ethics and Governance Team if you wish to make a complaint or comment please email cgr-reg@exeter.ac.uk,

You can also contact members of the research team:
Julian Dyer (j.dyer3@exeter.ac.uk)
Ellen Dyer (ellen.dyer@ouce.ox.ac.uk)
Dennis Ochieng (denniso@uonbi.ac.ke)
Peter Opeyemi James (pj324@exeter.ac.uk)
Dalmas Omia (ochiengomia@uonbi.ac.ke)

Thank you for your interest in this project.

CONSENT

1. I confirm that I have read the information provided above.

2. I confirm that I am 18 years old or older.

3. I understand that my participation is voluntary and that I am free to withdraw at any time.

4. I understand that relevant sections of the data collected during the study may be looked at by members of the research team and individuals from the University of Exeter where it is relevant to my taking part in this research. My responses will also be anonymously reviewed by a panel of experts. I give permission for these individuals to have access to my records.

6. I understand that taking part involves survey data to be collected and used for the purposes of anonymized inclusion in academic publications, project publication/website,

and further grant applications.

7. I agree to take part in the above project.

- ○ I confirm that I have read the consent statements and agree to participate in this study.
- ○ I do not consent to agree to participate in this study.

Block 1 - Recommended Policies

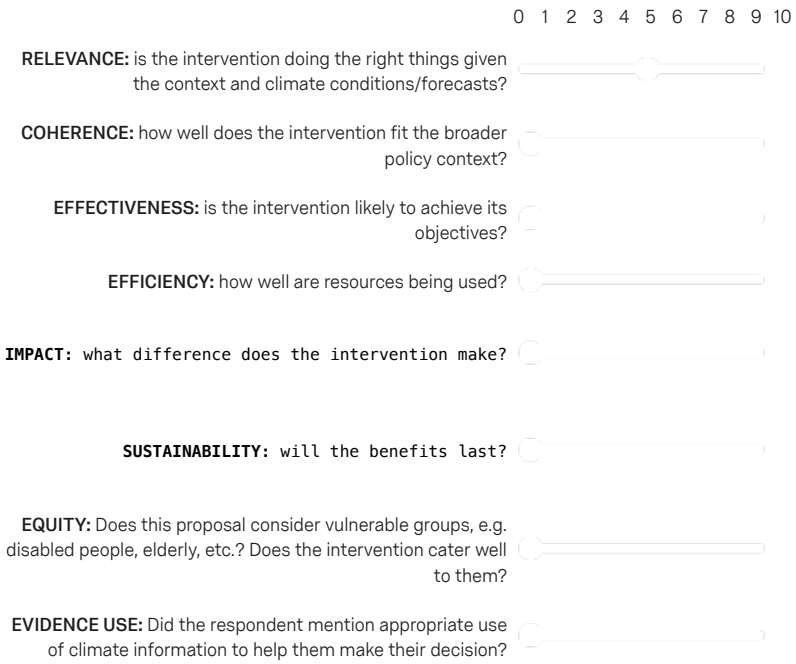
The first question asked was:

"If you had a budget of 500,000 KES to spend on preparation for climate hazards in your ward/community this season, what would you prioritize? Why?"

In response to this question, the respondent gave the following answer:

[PIPED TEXT - ANSWER GOES HERE]

Please rate the response on the following seven categories:



[OPTIONAL] Do you have any comments on your answer above? Are there considerations not reflected in the three criteria?

Block 2 - Choice Between Programs

In the next question, the respondent was asked:

"This season, an NGO approaches you and says they have either a feeding program or a livestock vaccination project but can only implement one in your area. Which would you prefer? Why?"

In response to this question, the respondent gave the following answer:

[PIPED TEXT - ANSWER GOES HERE]

Please rate the response on the following seven categories:

	0	1	2	3	4	5	6	7	8	9	10
RELEVANCE: is the intervention doing the right things given the context and climate conditions/ forecasts?											
EQUITY: Does this proposal consider vulnerable groups, e.g. disabled people, elderly, etc.? Does the intervention cater well to them?											
EVIDENCE USE: Did the respondent mention appropriate use of climate information to help them make their decision?											

[OPTIONAL] Do you have any comments on your answer above? Are there considerations not reflected in the seven criteria?

Block 3 - Scenario 1

The participant was then asked about this first hypothetical scenario:

Scenario 1: "It is late March and the long rains have not yet begun. People in your community are concerned that the season may fail and some have begun talking of long migrations into areas near the Pokot border and possibly into South Sudan." How would you respond to this scenario? Please be as detailed as possible, explain all the different resources you would draw on, and explain all the crucial decisions you make to arrive at your proposed policy plan for how the government should respond to this situation.

In response to this question, the respondent gave the following answer:

[PIPED TEXT - ANSWER GOES HERE]

Please rate the response on the following seven categories:

	0 1 2 3 4 5 6 7 8 9 10
RELEVANCE: is the intervention doing the right things given the context and climate conditions/forecasts?	
COHERENCE: how well does the intervention fit the broader policy context?	
EFFECTIVENESS: is the intervention likely to achieve its objectives?	
EFFICIENCY: how well are resources being used?	
IMPACT: what difference does the intervention make?	
SUSTAINABILITY: will the benefits last?	
EQUITY: Does this proposal consider vulnerable groups, e.g. disabled people, elderly, etc.? Does the intervention cater well to them?	
EVIDENCE USE: Did the respondent mention appropriate use of climate information to help them make their decision?	

[OPTIONAL] Do you have any comments on your answer above? Are there considerations not reflected in the three criteria?

Scenario 2:

“At a community baraza, a number of people mentioned that the river appears to be running unusually high for this time of year. They think there may be flooding this season and have brought their concerns to you.”

How do you think the county government should respond to this scenario? What would be the proposed response you suggest to the government? Please be as detailed as possible, explain all the different resources you would draw on, and explain all the crucial decisions you make to arrive at your proposed policy plan for how the government should respond to this situation.

Block 4 - Scenario 2

The participant was then asked about this first hypothetical scenario:

Scenario 2: "At a community baraza, a number of people mentioned that the river appears to be running unusually high for this time of year. They think there may be flooding this season and have brought their concerns to you." How do you think the county government should respond to this scenario? What would be the proposed response you suggest to the government? Please be as detailed as possible, explain all the different resources you would draw on, and explain all the crucial decisions you make to arrive at your proposed policy plan for how the government should respond to this situation.

In response to this question, the respondent gave the following answer:

[PIPED TEXT - ANSWER GOES HERE]

Please rate the response on the following seven categories:

	0 1 2 3 4 5 6 7 8 9 10
RELEVANCE: is the intervention doing the right things given the context and climate conditions/forecasts?	
COHERENCE: how well does the intervention fit the broader policy context?	
EFFECTIVENESS: is the intervention likely to achieve its objectives?	
EFFICIENCY: how well are resources being used?	
IMPACT: what difference does the intervention make?	
SUSTAINABILITY: will the benefits last?	
EQUITY: Does this proposal consider vulnerable groups, e.g. disabled people, elderly, etc.? Does the intervention cater well to them?	
EVIDENCE USE: Did the respondent mention appropriate use of climate information to help them make their decision?	

[OPTIONAL] Do you have any comments on your answer above? Are there considerations not reflected in the three criteria?

Feedback

Thank you for participating in this exercise. You have now finished the survey.

If you have any comments, questions, or feedback about the study or your participation, please share them in the box below and feel free to contact researchers by email:

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IRB / Ethical Approval